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# Image Processing

*Presentation of lectures*

Department of Computer Graphics and Multimedia



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*Lecture topic*

# **Introduction, image representation**

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*„What is images?  
What is it that we want to process?“*

# Contents:

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- Motivation and application
- Image representation
- Overview of image processing topics

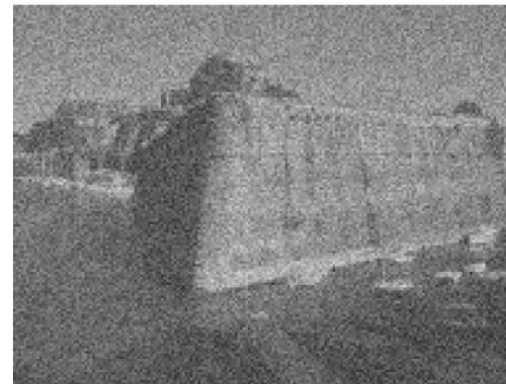
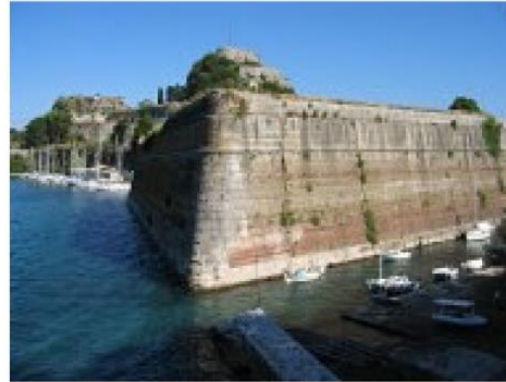
# Motivation and application

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- “Image processing” is a technological/research field that focuses processing and analysis of image content (unlike e.g. computer graphics that focuses synthesis)
- Image processing solved tasks, such as:
  - What to do with an image to get rid of noise?
  - How to enhance edges? How to improve contrast?
  - How to find human faces?
  - Where are object limits? Which parts belong together?
- Image processing is „followed“ by computer vision, a field that focuses “higher” analyses, such as image objects reconstruction, understanding of image contents, etc.

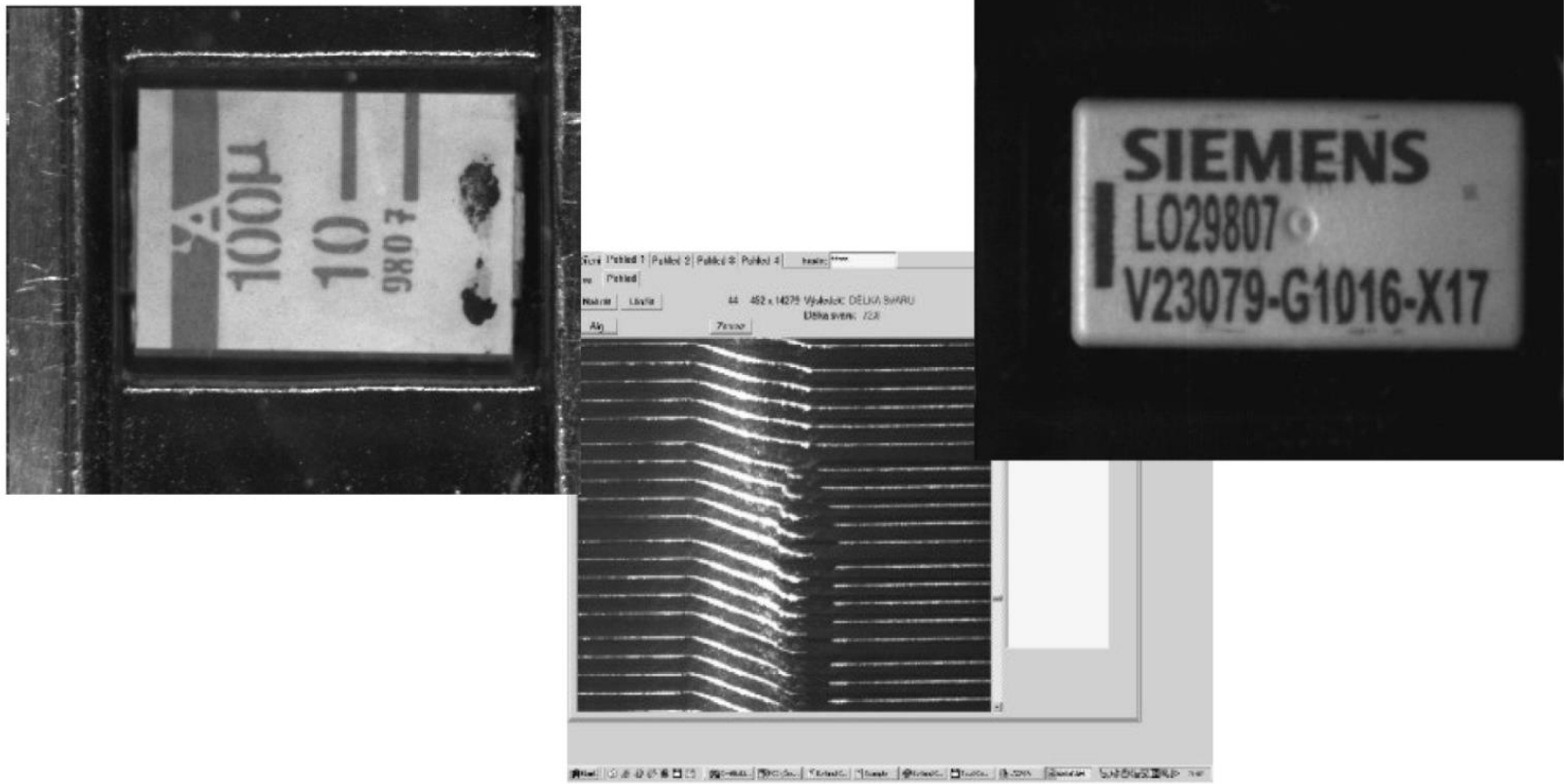
# Motivation and application – typical apps

- Preparation of image for publication (print) – adjustment of contrast, focus, etc.



# Motivation and application - industry

- Quality control in industry using camera systems with image processing (capacitors, welding processes, etc.)



# Motivation and application - traffic

- Monitoring of traffic, enforcement (profiles of vehicles, traffic sign detection, speed, red light, etc.)





# Image representation – function, set...

- Images can be seen as functions of 2 variables or 2D continuous signal, or otherwise...

- For example by a function  $I_{x,y} = f(x, y)$  g.  $I_{x,y} = \sin(x^2 + y)$

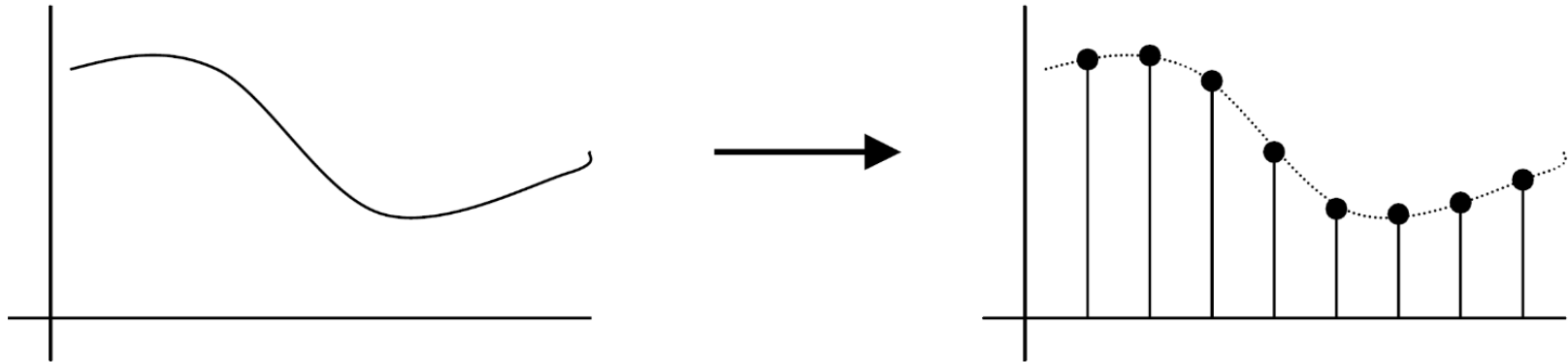
- ... by a set of points:  $I = \left\{ (1,2)(2.8)\left(3, \frac{7}{13}\right)(2.2, \sqrt{7})... \right\}$

- ... by an infinite set of points fulfilling some criteria:

$$I = \left\{ (x, y) \mid x^2 + y^2 = r^2 \right\}$$

# Image representation – discrete samples

- Contemporarily, images are typically represented by a raster of points Pixels (Picture elements), bitmap, array:
- The relationship of the function of 2 variables and the image raster is such that the raster image is a discrete representation of the function obtained by sampling

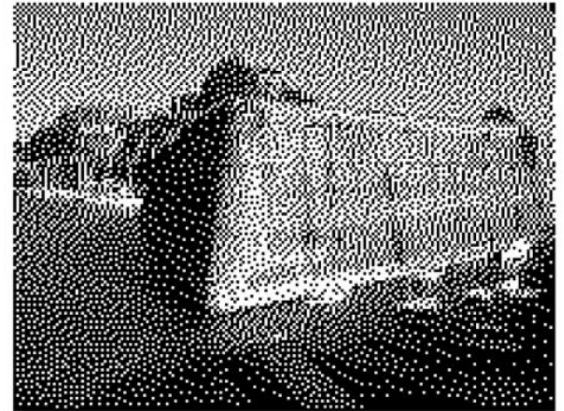


- Note that the function is correctly represented (no information loss if the sampling theorem is fulfilled (spatial frequencies, e.g. in )

$$f_{\max} < \frac{1}{2} f_s$$

# Image representation – functional values

- The image functional value is „naturally“ colour spectrum; however, in computers, we typically use suitable colour representation of grayscale (monochromatic) images
- Most of the information in images is contained in the luminance (intensity, brightness) of the image pixels, therefore, grayscale is often used (no colour)
- The functional values are sampled (discrete), we often use 0-255, 0-65535, or RGB (3x 0-255), but other are possible...



# Image representation – image size

- Typically, images are limited in size and its values are discrete, therefore, instead of the „idealistic“ representation:

$$I_{x,y} = f(x, y), f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R} \times \mathbb{R}$$

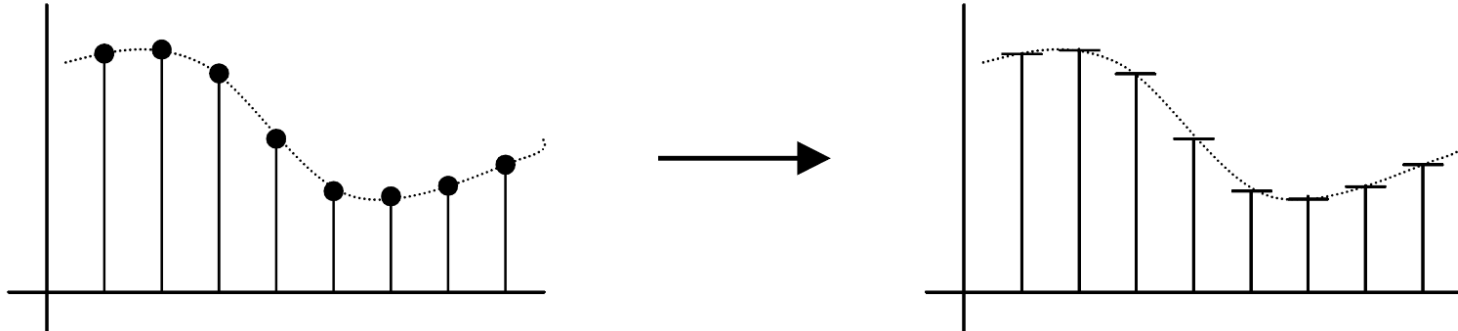
... we typically get the computer representable image, e.g.:

$$I_{x,y} = f(x, y), f : \langle 0, 1023 \rangle \times \langle 0, 1023 \rangle \rightarrow \langle 0, 255 \rangle \times \langle 0, 255 \rangle \times \langle 0, 255 \rangle$$

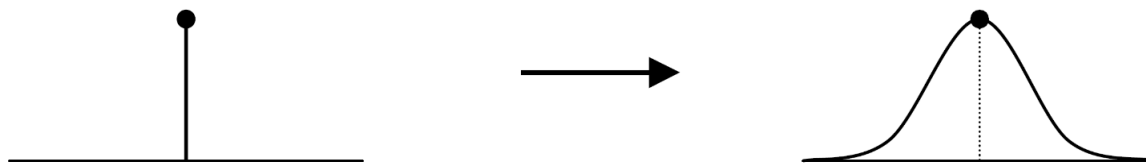
# Image representation – sampling reality

At a first glance, it seems that to fulfill the sampling theorem may be difficult; however, „helping“ factors exist:

- 1) The image sensors cannot perform „point“ sampling but they rather use small surfaces (areas).



- 2) The optical systems (lenses) are not ideal and they slightly (or significantly) „defocus“ (typically with Gaussian):



# Overview of image processing topics

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- Image data acquisition, distortion of image
- Point transformations (functions), colours
- 2D Fourier transform, 2D FFT, Wavelet transforms, DCT
- Linear filtrations, correlation, convolution
- Features of Fourier transform, fast correlation, convolution
- Geometrical features and transforms (rotation, scale, warp)
- Morphological filtering
- Edge detection, object detection, segmentations
- Image distortions, noise, optimal filtration
- Watermarks

# Conclusions

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- Image processing (and following computer vision) is a nice and perspective field with a wide application potential and it is also unexplored (computer vision)
- Representation of image is a key stone for understanding and exploration in image processing
- A typical representation of image is, nowadays, representation thorough a raster but alternatives exist
- Sampling theorem in image acquisition is not that difficult to fulfill as optical systems „help“
- Of course, more about images follows in next lectures